

Coronary Artery Disease and Heart Failure

1. Overview

Coronary artery disease (CAD) is the leading cause of death globally, responsible for approximately 9 million deaths per year. It results from atherosclerosis — progressive narrowing of coronary arteries by lipid-laden plaques.

Heart failure (HF) is a clinical syndrome where the heart cannot pump sufficient blood to meet the body's needs or can only do so at elevated filling pressures. CAD is the most common cause of heart failure, followed by hypertension and cardiomyopathy.

India has one of the highest burdens of CAD, with disease occurring a decade earlier than in Western populations. Approximately 45% of acute MI deaths in India occur in people under 55 years of age.

2. Pathophysiology of Atherosclerosis

Atherosclerosis begins with endothelial dysfunction triggered by risk factors such as dyslipidemia, hypertension, diabetes, and smoking. LDL particles enter the intima of arterial walls, oxidize, and are engulfed by macrophages, forming foam cells — the core of the early fatty streak.

Over decades, lipid-laden foam cells, smooth muscle cells, and fibrosis form a plaque covered by a fibrous cap. A stable plaque causes angina during exertion. A vulnerable plaque with a thin fibrous cap can rupture, exposing the lipid core and triggering platelet aggregation and thrombosis — causing acute coronary syndrome (ACS).

Acute Coronary Syndromes (ACS) include: Unstable Angina (UA — chest pain at rest or increasing frequency, no biomarker rise), NSTEMI (Non-ST elevation MI — biomarker rise without ST elevation), and STEMI (ST elevation MI — complete coronary occlusion, time-critical emergency).

3. Risk Factors for CAD

Major modifiable: hypercholesterolemia (especially elevated LDL and low HDL), hypertension, diabetes mellitus, smoking, obesity, physical inactivity, unhealthy diet, and metabolic syndrome.

Non-modifiable: age (men >45, women >55), male sex (women's risk increases post-menopause), family history of premature CAD (first-degree relative with MI before age 55 in men, 65 in women), and South Asian ethnicity.

Emerging risk factors: elevated Lp(a) lipoprotein, high sensitivity CRP (hsCRP), homocysteine, air pollution exposure (India's urban areas have among the highest particulate matter levels globally), chronic kidney disease, HIV, autoimmune diseases.

4. Clinical Presentations

Stable Angina: exertional chest pain (pressure, tightness, burning) in the substernal area, radiating to jaw, left arm, or back. Relieved by rest and sublingual nitrates within 1-5 minutes. Indicates significant but stable coronary stenosis.

Unstable Angina/NSTEMI: angina at rest, new-onset angina, or crescendo angina. Requires urgent hospitalization. ECG may show ST depression or T-wave changes. Troponin elevated in NSTEMI.

STEMI: sudden onset severe crushing chest pain, diaphoresis, nausea, vomiting, dyspnea. ECG shows ST elevation in ≥ 2 contiguous leads or new LBBB. This is a time-critical emergency — 'Door-to-Balloon' time should be <90 minutes.

Atypical presentations: women, elderly, and diabetics may present with atypical symptoms — fatigue, jaw pain, upper back pain, dyspnea, or nausea without chest pain. High clinical suspicion required.

5. Diagnosis and Investigations

Test	Purpose / Findings
ECG	ST changes, Q waves, LBBB, arrhythmias
Troponin I or T (high-sensitivity)	Myocardial necrosis marker — elevated in MI
Chest X-ray	Cardiomegaly, pulmonary edema, pleural effusion
Echocardiogram	Wall motion abnormalities, EF, valve disease

Exercise Stress Test (TMT)	Provoke ischemia in suspected stable CAD
CT Coronary Angiography	Non-invasive coronary anatomy, calcium scoring
Conventional Angiography (CAG)	Gold standard; guides PCI/CABG decision
BNP / NT-proBNP	Heart failure diagnosis and monitoring

6. Medical Treatment of CAD

Antiplatelet therapy: aspirin 75-100 mg daily (lifetime). After ACS: dual antiplatelet therapy (DAPT) — aspirin + ticagrelor (90 mg BD) or clopidogrel (75 mg OD) for 12 months for ACS; 1-6 months after elective PCI.

Statins: high-intensity statin (atorvastatin 40-80 mg or rosuvastatin 20-40 mg) for all CAD patients, regardless of baseline LDL. Target LDL <70 mg/dL (very high risk) or <55 mg/dL (extremely high risk, post-ACS). Consider ezetimibe or PCSK9 inhibitors if target not reached.

Beta-blockers: metoprolol, carvedilol, bisoprolol — reduce heart rate, BP, and myocardial oxygen demand. Mandatory after MI. Continue indefinitely in stable CAD, especially with LV dysfunction.

ACE inhibitors/ARBs: mandatory post-MI with reduced EF (<40%), diabetes, or hypertension. Reduce LV remodeling, mortality. Titrate to maximum tolerated dose.

Nitrates: sublingual GTN for acute angina relief. Long-acting nitrates (isosorbide mononitrate, isosorbide dinitrate) for angina prevention. Nitrate-free interval essential to prevent tolerance.

7. Revascularization

Percutaneous Coronary Intervention (PCI): balloon angioplasty + coronary stent. Primary PCI (pPCI) is the preferred reperfusion strategy for STEMI within 90 minutes of first medical contact. Drug-eluting stents (DES) reduce restenosis rates. Dual antiplatelet therapy required for 6-12 months post-stenting.

Coronary Artery Bypass Grafting (CABG): preferred for left main disease, 3-vessel disease, diabetes with multivessel disease, and when PCI is technically not feasible. Uses internal mammary artery (IMA) graft for best long-term patency (>90% at 10 years).

Thrombolysis (fibrinolytic therapy): used for STEMI when pPCI is not available within 120 minutes. Streptokinase (less expensive) or tenecteplase (single bolus, more convenient).

Should be administered within 12 hours of symptom onset. 'Pharmacoinvasive strategy' — thrombolyse then transfer for early angiography.

8. Heart Failure — Overview and Classification

Heart failure is classified by ejection fraction (EF): HFrEF (EF <40%), HFmrEF (EF 40-49%), HFpEF (EF ≥50%). HFrEF has the most evidence-based treatment. HFpEF (common in hypertensive, diabetic, elderly women) remains a therapeutic challenge.

NYHA Functional Classification: Class I (no symptoms), Class II (mild symptoms with ordinary activity), Class III (symptoms with less than ordinary activity), Class IV (symptoms at rest).

Common causes: CAD/post-MI (most common), hypertension, dilated cardiomyopathy (DCM), valvular heart disease (rheumatic heart disease in India), viral myocarditis, alcoholic cardiomyopathy, thyroid disease.

9. Heart Failure Treatment

Four pillars of HFrEF treatment ('Fantastic Four'): ACE inhibitor/ARB/ARNI (sacubitril-valsartan) + Beta-blocker (carvedilol/bisoprolol/metoprolol XL) + Mineralocorticoid antagonist (spironolactone/eplerenone) + SGLT2 inhibitor (dapagliflozin/empagliflozin).

Diuretics: furosemide (loop diuretic) for symptom relief of fluid overload — the single most important drug for symptom control. Titrate dose to achieve dry weight (target JVP normal, no peripheral edema, no orthopnea). Does not improve mortality.

Device therapy: ICD (implantable cardioverter-defibrillator) for EF <35% on optimal medical therapy, for primary prevention of sudden cardiac death. CRT (cardiac resynchronization therapy) for EF <35% with LBBB and NYHA III-IV.

10. Cardiac Emergencies — What to Do

STEMI: call 112/911 immediately. Give aspirin 300 mg to chew (if not already taking, not allergic, not bleeding). Keep patient at rest, calm, monitor consciousness. No food or water if consciousness deteriorating. Transport to PCI-capable centre urgently. Time = muscle: every 30 minutes of delay increases 1-year mortality by ~7.5%.

Cardiac Arrest: shout for help. Call 112. Start CPR: 30 compressions (centre of chest, depth 5-6 cm, rate 100-120/min) then 2 rescue breaths. Minimize interruptions. Use AED as soon as available — turn on, follow voice prompts, deliver shock if advised. Continue CPR until AED says no shock needed and patient starts breathing, or until emergency services take over.

Acute Pulmonary Oedema: sit patient up (legs dangling). Call 112. Give supplemental oxygen. IV furosemide 40-80 mg (in hospital). IV nitrates (nitroglycerin infusion). Non-invasive ventilation (CPAP/BiPAP) if SpO₂ does not improve.